

Disk, disk, sleuth! II

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

All we know is the file with the flag is named **down-at-the-bottom.txt...**

Hints

- The sleuthkit has some great tools for this challenge as well.
 - Sleuthkit docs here are so helpful: TSK Tool Overview
 - This disk can also be booted with qemu!
-

Tools Used

- `7z` – Extract compressed files
 - `file` – Determine file type
 - `strings` – Extract printable character sequences
 - `exiftool` – Read metadata
 - `fdisk` – View partition tables
 - `srch_strings` – Search for strings in binary files (Sleuth Kit)
 - `fls` – List files in a disk image (Sleuth Kit)
 - `icat` – Retrieve a file by inode number (Sleuth Kit)
-

Complete Solution

Step 1: Download and Extract

Download the disk image named `dds2-alpine.flag.img.gz` from the challenge page.

Extract the compressed disk image:

```
7z x dds2-alpine.flag.img.gz
```

This decompresses the gzipped file and creates the raw disk image `dds2-alpine.flag.img`.

Step 2: Basic File Inspection

Verify the file type and understand its structure:

```
file dds2-alpine.flag.img
```

Expected output:

```
dds2-alpine.flag.img: DOS/MBR boot sector; partition 1 : ID=0x83, active, start-CHS  
(0x0,32,33), end-CHS (0x10,81,1), startsector 2048, 260096 sectors
```

The output confirms this is a **DOS/MBR boot sector** with a Linux partition (ID `0x83`).

Step 3: Search for Visible Text

A quick `strings` scan to see if the flag is hiding in plain sight:

```
strings dds2-alpine.flag.img | grep -i "picoCTF"
```

Output: (no results)

No luck. The flag isn't directly visible.

Step 4: Inspect Metadata

Examine the disk image metadata:

```
exiftool dds2-alpine.flag.img
```

Expected output:

```
ExifTool Version Number      : 13.50
File Name                    : dds2-alpine.flag.img
Directory                   : .
File Size                    : 134 MB
File Modification Date/Time  : 2025:12:19 20:45:43+02:00
...
Error                        : Unknown file type
```

Nothing suspicious here.

Step 5: Search for the Target Filename

We know the flag file is named `down-at-the-bottom.txt`. Let's use `srch_strings` (from The Sleuth Kit) to locate it:

```
srch_strings -a dds2-alpine.flag.img | grep 'down-at-the-bottom.txt'
```

Output:

```
down-at-the-bottom.txt
down-at-the-bottom.txt
```

Key finding: The filename appears twice in the disk image. This confirms the file exists and we can find it!

Step 6: View Partitions with `fdisk`

Use `fdisk` to see the partition structure in detail:

```
fdisk -l dds2-alpine.flag.img
```

Expected output:

Disk dds2-alpine.flag.img: 128 MiB, 134217728 bytes, 262144 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x5d8b75fc

Device	Boot	Start	End	Sectors	Size	Id	Type
dds2-alpine.flag.img1	*	2048	262143	260096	127M	83	Linux

The partition starts at sector **2048** and has **260096** sectors.

Step 7: List Files with fls (Sleuth Kit)

Using the offset `2048` (start of the partition), list all files and directories:

```
fls -o 2048 dds2-alpine.flag.img
```

Output:

```
d/d 26417:      home
d/d 11: lost+found
r/r 12: .dockerenv
d/d 20321:      bin
d/d 4065:       boot
d/d 6097:       dev
d/d 2033:       etc
d/d 8129:       lib
d/d 14225:      media
d/d 16257:      mnt
d/d 18289:      opt
d/d 16258:      proc
d/d 18290:      root
d/d 16259:      run
d/d 18292:     /sbin
d/d 12222:      srv
d/d 16260:      sys
d/d 18369:      tmp
d/d 12223:      usr
d/d 14229:      var
V/V 32513:      $OrphanFiles
```

The `root` directory (inode `18290`) looks promising.

Step 8: List Contents of the root Directory

```
fls -o 2048 dds2-alpine.flag.img 18290
```

Output:

```
r/r 18291:      down-at-the-bottom.txt
```

Bingo! The file we're looking for has inode `18291` .

Step 9: Extract the File with icat

Use `icat` to retrieve the file contents by its inode number:

```
icat -o 2048 dds2-alpine.flag.img 18291
```

Output:

```
  _  _  _  _  _  _  _  _  _  _  _  _  _  
/ \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \  
( p ) ( i ) ( c ) ( o ) ( C ) ( T ) ( F ) ( { ) ( f ) ( 0 ) ( r ) ( 3 ) ( n )  
 \ /  \ /  \ /  \ /  \ /  \ /  \ /  \ /  \ /  \ /  \ /  
  _  _  _  _  _  _  _  _  _  _  _  _  _  
/ \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \  
( s ) ( 1 ) ( c ) ( 4 ) ( t ) ( 0 ) ( r ) ( _ ) ( n ) ( 0 ) ( v ) ( 1 ) ( c )  
 \ /  \ /  \ /  \ /  \ /  \ /  \ /  \ /  \ /  \ /  \ /  
  _  _  _  _  _  _  _  _  _  _  _  
/ \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \  
( 3 ) ( _ ) ( 4 ) ( b ) ( d ) ( 7 ) ( 2 ) ( 1 ) ( f ) ( 2 ) ( } )  
 \ /  \ /  \ /  \ /  \ /  \ /  \ /  \ /  \ /  \ /  \ /
```

The flag is presented as ASCII art! Each character is wrapped in parentheses and placed between slashes.

Step 10: Read the Flag

Reading the characters from the ASCII art in order:

```
p i c o C T F { <REDACTED> }
```

Removing spaces gives the complete flag.

 Final Result

```
picoCTF{<REDACTED>}
```

Note: The actual flag is redacted here. In the real challenge, running the `icat` command will display the ASCII art with the complete flag.

 Key Concepts

Concept	Description
Raw Disk Image	A sector-by-sector copy of a storage device, preserving partitions and filesystems.
MBR (Master Boot Record)	The first sector of a disk that contains the partition table and boot code.
Linux Partition (0x83)	The partition type identifier for a native Linux filesystem (e.g., ext4).
fls	Lists files and directories in a disk image, showing inode numbers.
icat	Retrieves a file by its inode number without mounting the filesystem.
srch_strings	Sleuth Kit version of <code>strings</code> with additional forensic options.
ASCII Art	Text represented as graphical characters; here, the flag is hidden in this visual format.

Pro Tips

1. **Search for the filename first** – When you know the name of the target file, use `srch_strings` to quickly locate it in the disk image.
2. **`fls` reveals the filesystem structure** – Use it to explore directories and find inode numbers.
3. **`icat` is your best friend** – Extract files without mounting the filesystem, even if they're in weird formats.
4. **Don't ignore ASCII art** – Sometimes flags are presented in visually creative ways. Look for patterns like parentheses or slashes.
5. **Use `-o` to specify partition offset** – This tells Sleuth Kit tools where the filesystem starts within the disk image.
6. **`srch_strings -a` scans everything** – The `-a` option ensures you don't miss any strings, even those outside default data sections.